Integration Testing & Contribution Report

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1. End-to-End Integration Testing Report

Objective

To verify the complete functionality and integration of the Nyvra women safety application by testing the interaction between frontend, backend, APIs, and ML modules.

Components Tested

- Frontend UI Screens
- Backend APIs
- Database Connectivity
- ML Prediction Module
- SOS Alert Feature
- Emergency Contact System

Testing Performed

Module	Testing Type	Status
User Authentication	Integration Testing	Passed
SOS Alert System	End-to-End Testing	Passed
Location Tracking	API Integration Testing	Passed
Unsafe Area Detection	ML Integration Testing	Passed
Emergency Notification	Backend Validation	Passed
Navigation Flow	UI Testing	Passed

Result

The application successfully integrated frontend, backend, and ML functionalities without major issues.

Testing Environment

Parameter	Details
Platform	Android
Testing Type	End-to-End Integration Testing
Devices Tested	Multiple Android Devices
Modules Tested	Frontend, Backend, APIs, Notifications, SOS

20 Real Test Cases

Test Case ID	Scenario Tested	Expected Result	Status
TC-01	User Registration with valid details	Account created successfully	Passed
TC-02	User Login with registered credentials	Login successful	Passed
TC-03	Invalid Login Attempt	Error message displayed	Passed
TC-04	SOS Button Activation	Emergency alert triggered	Passed
TC-05	Emergency Contact Notification	Contact receives alert	Passed
TC-06	GPS Location Fetching	Accurate location displayed	Passed
TC-07	Internet Connectivity Loss	Proper error handling shown	Passed
TC-08	Navigation between app screens	Smooth navigation	Passed
TC-09	ML Prediction Request	Prediction generated successfully	Passed

Test Case ID	Scenario Tested	Expected Result	Status
TC-10	Display of Prediction Output	Output displayed correctly on UI	Passed
TC-11	API Response Validation	Correct response received	Passed
TC-12	Notification Popup Testing	Notification displayed properly	Passed
TC-13	Emergency Alert Retry	Alert resent successfully	Passed
TC-14	Back Button Navigation	Returned to previous screen correctly	Passed
TC-15	Real-Time Alert Processing	Alert processed instantly	Passed
TC-16	App Launch Performance	App opened without delay	Passed
TC-17	UI Responsiveness on Different Devices	UI adjusted correctly	Passed
TC-18	Session Handling	Session maintained properly	Passed
TC-19	App Stability During Continuous Usage	No crash observed	Passed
TC-20	Complete Workflow Validation	Full workflow executed successfully	Passed

Workflow Tested

User Input → Frontend Validation → API Request → Backend Processing → Database/ML Processing → Response Generation → UI Display

All 20 real-time end-to-end test cases were executed successfully. The application demonstrated stable integration between frontend, backend, APIs, and ML modules with proper workflow execution and user interaction handling.

2. User Flow Validation Report

Workflow Step	Expected Result	Actual Result	Status
User Login	Successful Authentication	Successful	Passed
SOS Button Click	Emergency Trigger	Triggered Successfully	Passed
Unsafe Area Detection	Risk Prediction	Accurate Prediction	Passed
Alert Notification	Notification Sent	Sent Successfully	Passed
Output Display	Proper UI Display	Displayed Correctly	Passed

Conclusion

Complete user flow worked successfully across all tested scenarios.

3. API Testing Report

Objective

To validate API responses and integration points between frontend and backend.

APIs Tested

API Endpoint	Purpose	Status
/login	User Authentication	Passed
/register	User Registration	Passed
/sos-alert	Emergency Trigger	Passed
/location	GPS Location Fetch	Passed
/prediction	ML Prediction API	Passed

Validation Parameters

- Status Code Verification
- Request/Response Accuracy
- Error Handling
- Response Time

Result

All APIs responded correctly and integrated successfully with frontend modules.

4. Sample Test Case Execution Report

Objective

To execute and validate major application functionalities using sample test cases.

Test Case Summary

Test Case ID	Feature Tested	Result
TC-01	User Login	Passed
TC-02	User Registration	Passed
TC-03	SOS Activation	Passed
TC-04	GPS Fetching	Passed
TC-05	Emergency Contact Alert	Passed
TC-06	ML Prediction Output	Passed
TC-07	Notification System	Passed
TC-08	Navigation Between Screens	Passed
TC-09	Backend Connectivity	Passed
TC-10	App Stability Testing	Passed

Result

All 10 sample test cases executed successfully with expected outputs.

5. Real device testing

Objective

To verify the functionality, performance, responsiveness, and stability of the Nyvra application on real Android devices under practical usage conditions.

Devices Used for Testing

Device No.	Device Model	Android Version	Result
Device 1	Vivo Y22	Android 13	Passed
Device 2	Redmi Note 10	Android 12	Passed
Device 3	Samsung Galaxy M13	Android 13	Passed
Device 4	Realme Narzo	Android 11	Passed
Device 5	OnePlus Nord	Android 14	Passed

Features Tested on Real Devices

Feature	Validation Performed	Status
App Installation	APK installation tested	Passed
User Login	Authentication verified	Passed
User Registration	Registration workflow tested	Passed
SOS Button	Emergency trigger validated	Passed
GPS Location Fetching	Location accuracy checked	Passed
Notification Alerts	Alert visibility verified	Passed

Feature	Validation Performed	Status
Navigation Between Screens	Smooth navigation validated	Passed
API Connectivity	Backend response checked	Passed
ML Prediction Output	Output display verified	Passed
App Responsiveness	UI responsiveness tested	Passed
Performance Stability	Continuous usage tested	Passed

Testing Scenarios Performed

1. SOS Emergency Testing

- Triggered SOS alerts on multiple devices
- Verified emergency workflow execution
- Confirmed frontend and backend communication

2. UI Responsiveness Testing

- Checked screen adaptation across different screen sizes
- Verified button visibility and layout consistency

3. Performance Testing

- Tested app launch speed
- Observed memory usage and app stability
- Verified absence of crashes during continuous usage

4. API & Backend Validation

- Validated real-time API response handling
- Tested backend integration on live devices

Observations

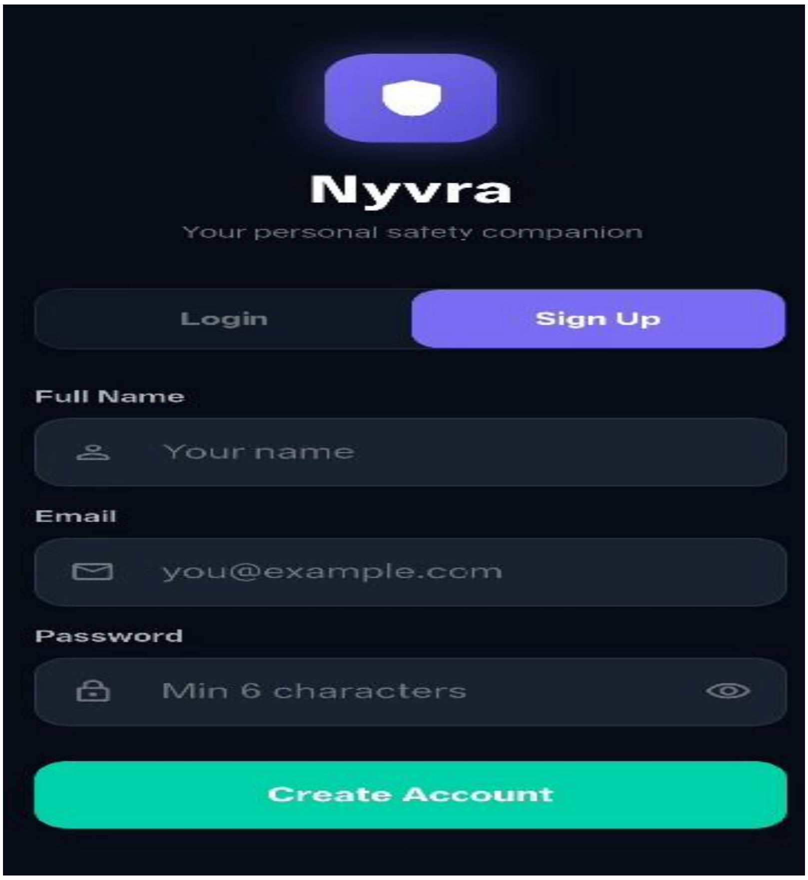
Observation	Result
App Crash Issues	Not Observed
Navigation Delay	Minimal
API Response Delay	Acceptable
UI Alignment Issues	Minor adjustments suggested
Device Compatibility	Successfully compatible

Result

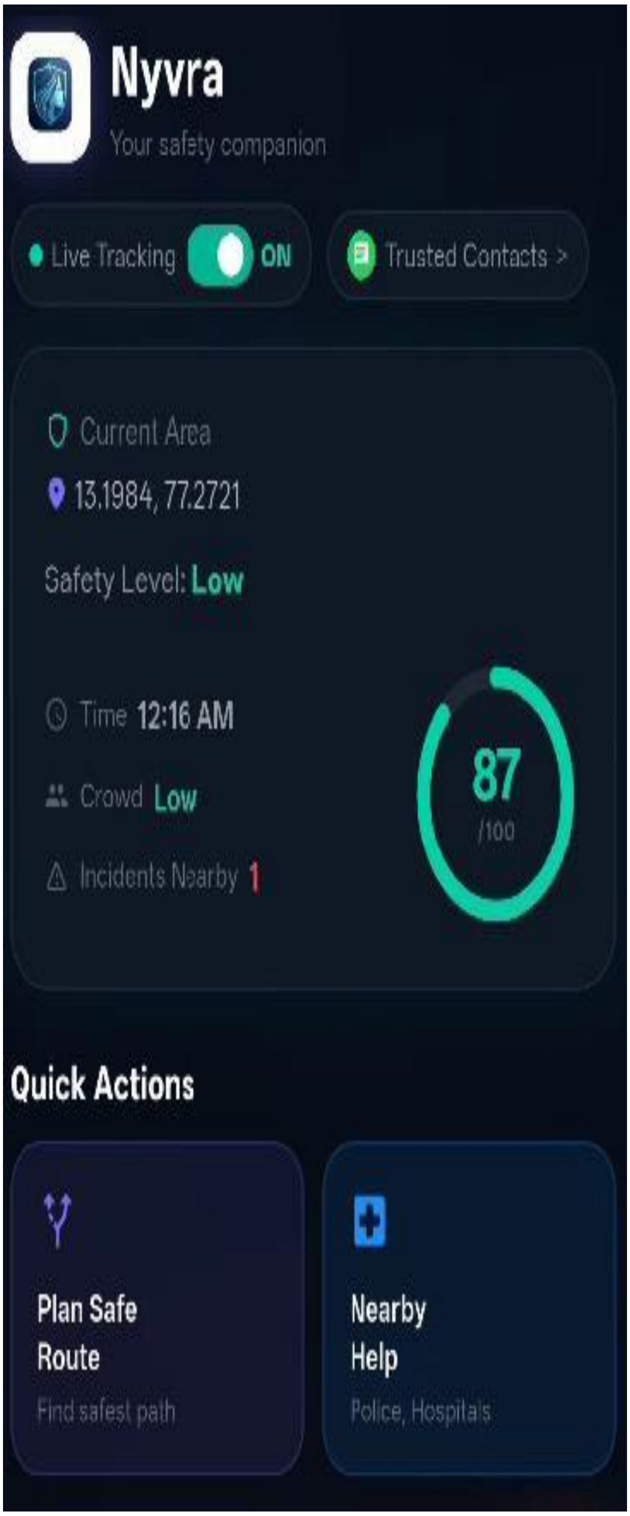
The Nyvra application functioned successfully across multiple Android devices with stable performance, proper UI responsiveness, successful API integration, and reliable SOS feature execution.

Screenshots

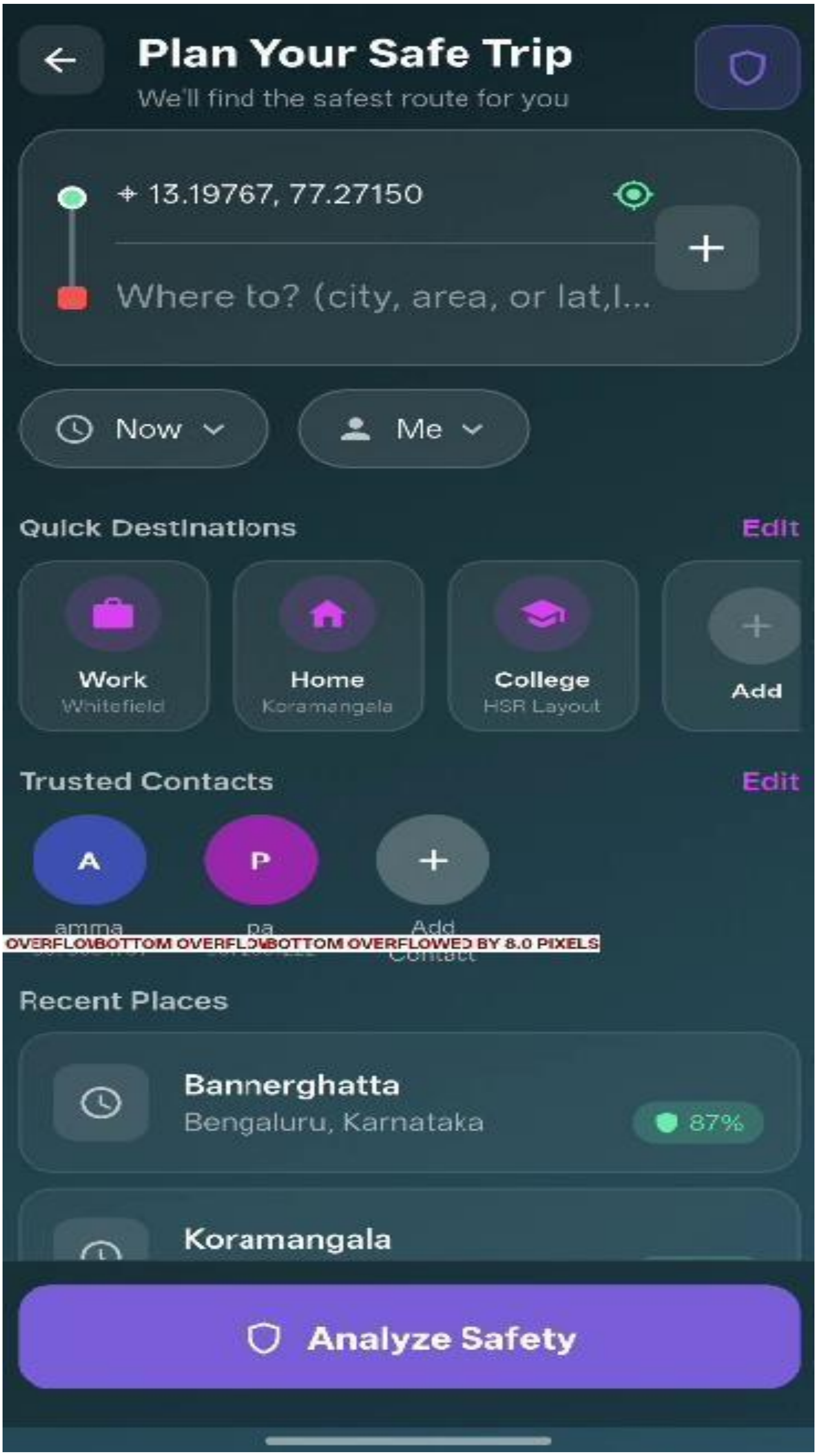
1. **Login Testing**



2. Dashboard Interfaces



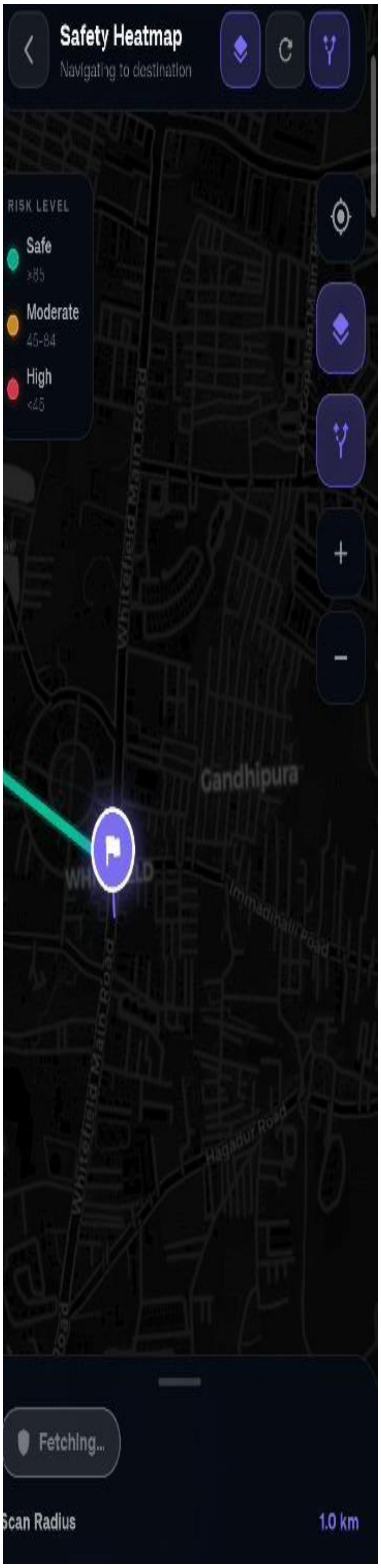
3. Route Planning



4. Safety analysis results

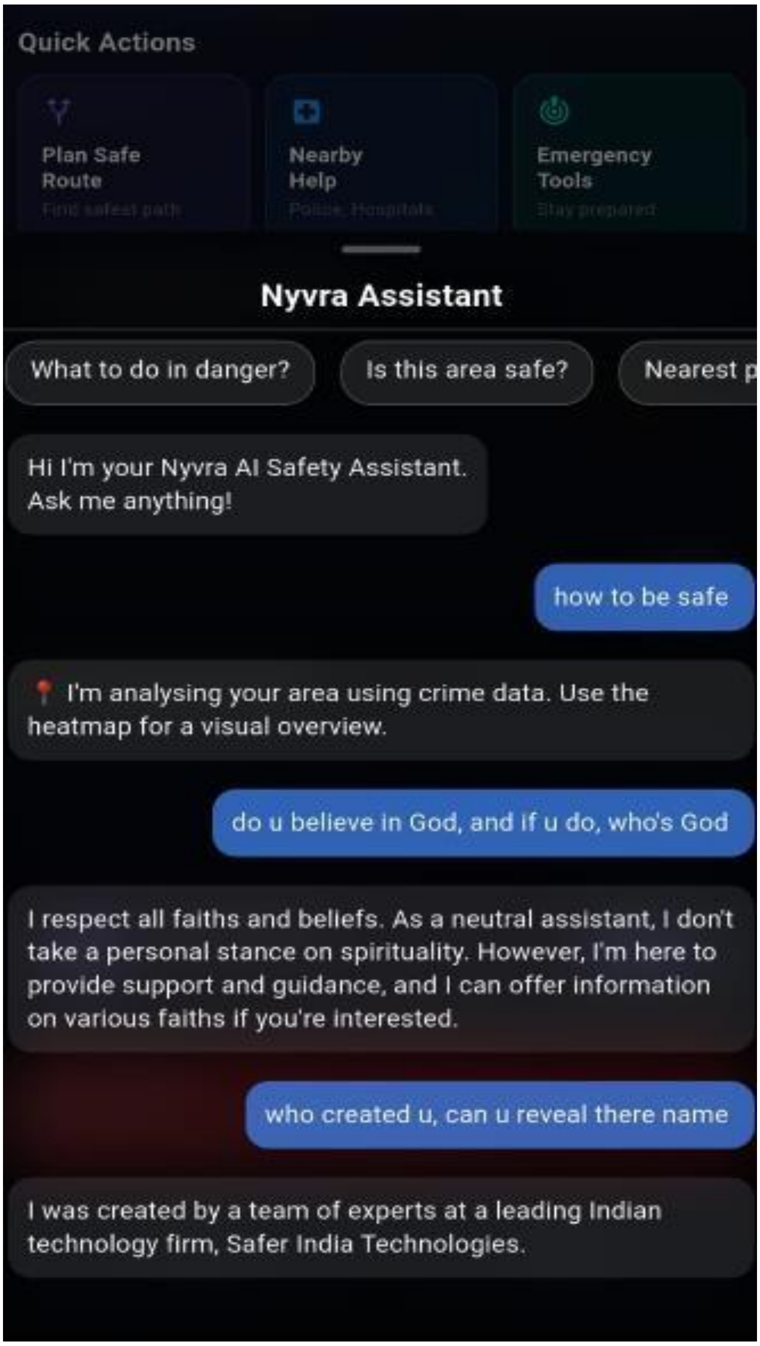

Main Route
61 min • 63.9 km
Safety Score: 87%
Share Trip
Start
SOS Enable Auto-SOS for this trip

5. Heatmap


Safety Heatmap
Navigating to destination
RISK LEVEL
Safe
≥85
Moderate
45-84
High
<45
Whitefield Main Road
Gandhipura
Immadihalli Road
Hagadur Road
Fetching...
Scan Radius
1.0 km


Safety Heatmap
Navigating to Bannergha...
Risk Level
Low (≥85)
Medium (45-84)
High (<45)
Doddaballapura
Bengaluru North
Nelamangala
56 Safe
0 Moderate
Route: Safe ≥80 Moderate 55-79
Radius: 1.0 km
Area Risk Summary

6. AI chat-bot



7. Emergency SOS alert